BIOL 250 — Systemic Quick-Reference Notes

Open-note speed reference — built from all 7 Systemic GR worksheets (Introduction, Cell, Tissue & Embryology, Integument, Osseous Tissue & Structure, Skeletal Muscle & Tissue, Skeletal) + Martini textbook context. All figure-labeling questions omitted.

QUICK-SCAN INDEX

Alphabetical fast-lookup — for use under exam time pressure (15-25 sec/question). Cross-references cell biology, tissue/embryology, integument, osseous, skeletal muscle, and skeletal sections.

Quick-Scan A–G

Term	One-line fact
Achondroplasia	Disproportionate short stature from slowed epiphyseal cartilage growth
Actinin	Protein forming the Z line meshwork, anchors thin filaments
Appositional cartilage growth	Perichondrium stem cells → chondroblasts add matrix at surface
Arthrocentesis	Joint fluid aspiration for diagnosis
Basal lamina	Glycoprotein/proteoglycan layer anchoring epithelium to connective tissue
Bone mass decline	Age-related decline typically begins around age 30-40
Cartilage (general)	Avascular; nutrients diffuse through matrix, heals slowly
Centriole	One of two structures within a centrosome; organizes microtubules
Centrosome	Region of cytoplasm containing the pair of centrioles
Cilia	Many short motile structures per cell; move fluid/mucus
Compact vs spongy bone	Compact = osteons; spongy = trabeculae, no osteons
DEXA scan	Bone density measurement
DNA replication	Helicase unwinds helix → polymerase synthesizes strand; occurs in S phase
Dermal blood supply with age	DECREASES with age
Diaphysis	Shaft of a long bone
Endocrine gland formation	Forms when connecting duct to epithelial surface disappears
Endochondral ossification	Bone replaces a hyaline cartilage model
Endomysium	Surrounds each individual muscle fiber
Epimysium	Surrounds the entire muscle; continuous with deep fascia
Epiphysis	Expanded end of a long bone
Flagellum	Whip-like motile structure, e.g. sperm tail; one per cell (vs many cilia)
Folliculitis	Inflammation/infection of a hair follicle
Gap junctions in cardiac muscle	Direct ion flow between cells → coordinated contraction
Gigantism	Excess GH before epiphyseal closure → excessive long bone growth
Golgi apparatus	Packages/secretes proteins AND renews plasmalemma via vesicle trafficking
Glycocalyx	Carbohydrate coating on cell surface — identity/recognition
Gap junctions (general)	Channel proteins, direct ion/small-molecule passage between cells
Greenstick fracture	Incomplete break, common in children
Comminuted fracture	Bone shattered into 3+ pieces
Compound (open) fracture	Bone breaks through the skin
Chondroblasts	Immature cartilage-forming cells
Chondrocytes	Mature cartilage cells, sit in lacunae
Collagen fibers	Triple-helix protein, high tensile strength
CAMs	Cell adhesion molecules — bind cells to each other and ECM
Calcitonin	Lowers blood calcium; stimulates osteoblasts/inhibits osteoclasts
Cyanosis	Bluish skin from low oxygenation
Carotene	Orange-yellow dietary pigment contributing to skin color
Desmosomes	Spot-welds linking adjacent cells via intermediate filaments
Diarthrosis	Freely movable synovial joint

Quick-Scan H–M

Term	One-line fact
Hematopoiesis	Blood cell formation, occurs in red marrow
Interphase	Phase in which a cell spends the MAJORITY of its life cycle
Interphase — G1 duration	Variable; can be brief or very long depending on cell type
Lacunae	Small cavities housing osteocytes
Lamellae	Layers of bone matrix (concentric, circumferential, interstitial)
Marfan's syndrome	Connective tissue disorder; abnormally long limbs
Melanosomes	Membrane-bound vesicles storing/transferring melanin
Mitochondria	DOUBLE membrane: outer + inner (cristae-folded); ATP production
Motor unit	One motor neuron + all fibers it controls
Myoglobin	O2-binding pigment giving slow fibers their red color
Hemidesmosomes	Anchor epithelial cells to underlying basal lamina
Histones	Proteins around which DNA winds to form nucleosomes
Hyaline cartilage	Resists compression; joint surfaces, costal cartilage
Hypertrophy	Increase in muscle fiber size from repeated near-max stimulation
Atrophy	Decrease in muscle size/fiber diameter from disuse
Intramembranous ossification	Bone forms directly from mesenchymal tissue, e.g. flat skull bones
Lysosomes	Digestive enzymes; breakdown debris/organelles/pathogens
Langerhans cells	Phagocytic immune cells, chiefly in stratum spinosum
Merkel cells	Sensory touch cells, found in stratum basale
Mast cells	Release histamine/heparin during inflammation
Meniscus	Fibrocartilage pad subdividing a synovial cavity

Quick-Scan N–S

Term	One-line fact
Osteoblasts	Secrete organic bone matrix (osteoid)
Osteoclasts	Multinucleated; resorb/dissolve bone matrix
Osteocytes	Mature bone cells, maintain matrix, sit in lacunae
Osteomalacia	Adult bone softening from vitamin D deficiency / poor mineralization
Osteon	Structural unit of compact bone, concentric lamellae around Haversian canal
Peroxisomes	Oxidase/catalase enzymes break down fatty acids, toxins (e.g. H2O2, alcohol)
Plain X-ray	Standard imaging for bone/joint structure
Race/ancestry from skeleton	Forensic anthropologists CAN estimate this from skeletal features
Receptor-mediated endocytosis	e.g. iron (transferrin-bound), cholesterol (LDL-bound)
Red marrow	Hematopoietic marrow
Rickets	Childhood osteomalacia; impairs bone growth
Skeletal muscle count	Approximately 700 skeletal muscles in the human body
Stem cells in skin	Persist in BOTH epidermis (basale) AND dermis throughout life
Synaptic cleft	GAP between neuron and muscle fiber
Synaptic vesicle	Sac storing/releasing neurotransmitter — NOT the same as synaptic cleft
Synovial membrane lining	NOT true epithelium — derives from fibroblasts/macrophage-like cells
Nucleolus	Site of ribosomal RNA synthesis/ribosome assembly
Nucleosome	DNA wound around histone core — chromatin packaging unit
Osteoporosis	Excess bone resorption vs formation, common in older women
Osteoprogenitor cells	Stem cells that differentiate into osteoblasts after fracture
Periosteum	Outer covering of bone; fibrous + osteogenic layers
Perichondrium	Fibrous covering around cartilage (except articular surfaces)
Pinocytosis	"Cell drinking" — nonspecific uptake of extracellular fluid
Phagocytosis	"Cell eating" — engulfing large particles/pathogens

Term	One-line fact
Sebaceous glands	Holocrine secretion of sebum, linked to hair follicles
Sarcomere — A band	Dark band; full length of thick filaments, constant length
Sarcomere — H band	Region of A band with thick filaments only, no overlap
Sarcomere — I band	Light band; thin filaments only
Subluxation	Partial dislocation
Sesamoid bone	Small bone embedded in a tendon, e.g. patella

Quick-Scan T–Z

Term	One-line fact
Tissue aging (general)	Reduces regenerative capacity, vascularity, elasticity across systems
Tonofibrils	Keratin precursor intermediate-filament bundles in keratinocytes
Trabeculae	Struts of spongy bone, no osteons
Wrinkle causes	Dermal thinning, hormonal change, UV exposure, pregnancy/weight change
Yellow marrow	Dominated by adipocytes (fat cells)
Z line	Marks end of sarcomere; anchors thin filaments via actinin
Titin	Elastic protein connecting Z line to thick filament/M line
Troponin	Binds Ca2+, shifts tropomyosin to expose actin active sites
Tropomyosin	Covers myosin-binding sites on actin at rest
Triad	One T tubule + two terminal cisternae of SR
T tubule	Sarcolemma extension distributing the action potential
Transitional epithelium	Stratified; cells change shape to allow distention, urinary tract
Vellus hair	Fine, pale "peach fuzz" hair
Terminal hair	Coarse, pigmented hair (scalp, eyebrows, post-puberty axilla/pubic)
Zygote	Fertilized egg, single cell formed at conception
Blastocyst	Early hollow ball-of-cells stage of embryo

1. Anatomical Organization & Terminology

1.1 Levels of organization (lowest → highest)

Level	Description
Chemical / molecular	Atoms interact to form molecules with specific properties (e.g., proteins, DNA)
Cellular	Molecules organized into organelles, then cells — smallest living units
Tissue	Groups of cells + cell products working together (epithelial, connective, muscle, neural)
Organ	Combination of two+ tissue types performing specific, complex function
Organ system	Organs interacting to perform a common, broad set of functions
Organism	All organ systems working together; the highest level of complexity

Order from highest to lowest complexity (per GR phrasing): organism → organ system → organ → tissue → cellular → chemical/molecular.

1.2 Anatomical subdisciplines

Term	Definition
Cytology	Study of the structure/function of individual cells — smallest units of life
Histology	Study of tissues — groups of specialized cells + products working together
Embryology	Study of early developmental processes (conception → ~2 months)
Developmental anatomy	Study of structural changes from conception through physical maturity
Gross anatomy	Study of structures visible to the naked eye; includes surface, regional, systemic
Surface anatomy	Study of general form/morphology and superficial markings
Regional anatomy	Study of all superficial + internal features in one specific body area
Systemic anatomy	Study of structure of one particular organ system (e.g., cardiovascular)
Comparative anatomy	Study of anatomical structures across different species
Clinical anatomy	Study of anatomical features that change recognizably during illness

Term	Definition
Radiographic anatomy	Study of structures via imaging (CT, MRI, ultrasound, angiography) — newer subspecialty of gross anatomy enabled by CT/spiral scan advances
Surgical anatomy	Anatomy relevant to surgical technique/landmarks
Cross-sectional anatomy	Anatomy as seen in cross-sectional images/slices
Physiology	Study of function (vs. anatomy = study of structure)

Gross anatomy specialties = radiographic anatomy + surgical anatomy + cytology + embryology + histology (per GR Q38 framing, though cytology/histology/embryology are technically microscopic, not gross — know the GR's grouping).

1.3 Directional terms & body planes

Term	Meaning
Lateral	Away from the midline
Medial	Toward the midline
Proximal / distal	Closer to / farther from point of attachment or trunk
Cephalic ↔ caudal	Toward the head ↔ toward the tail/inferior end (anatomical equivalents pair)
Anterior ↔ ventral; posterior ↔ dorsal	Front ↔ front; back ↔ back (equivalent pairs)
Superficial ↔ deep	Closer to ↔ farther from the body surface
Coronal (frontal) section	Vertical plane dividing body into anterior + posterior portions
Transverse (cross-sectional) section	Horizontal plane dividing body into superior + inferior portions; at right angles to long axis
Sagittal section	Vertical plane dividing body into left + right portions
Midsagittal section	Sagittal section exactly through the midline → equal left/right halves (e.g., through the umbilicus)
Parasagittal section	Sagittal section offset from midline → unequal left/right portions

1.4 Body position & cavities

Term	Meaning
Anatomical position	Standing, feet flat together, eyes forward, hands at sides with palms facing anteriorly
Supine	Lying face up
Prone	Lying face down
Mediastinum	Central thoracic space containing heart, great vessels, esophagus, trachea (between the two pleural cavities)
Coelom	Embryonic body cavity that gives rise to thoracic + abdominopelvic cavities; the cavity containing organs of respiratory, cardiovascular, digestive, urinary, and reproductive systems collectively
Pleural cavity	Space surrounding each lung
Pelvic cavity	Inferior subdivision of abdominopelvic cavity
Serous membrane — visceral layer	Covers the surface of an internal organ
Serous membrane — parietal layer	Lines the wall of the cavity that contains the organ

1.5 Basic life processes & organ systems

Process	Definition
Metabolism	Sum of all chemical operations in the body (anabolism + catabolism)
Responsiveness (irritability)	Ability to act/respond to changes in the immediate environment
Growth	Increase in cell size and/or cell number
Differentiation	Cellular specialization to perform particular functions during development
Reproduction	Production of new cells or a new organism
Movement	Transport of materials (food, blood, etc.) within the body, or locomotion of the whole body
Excretion	Elimination of waste products / unneeded or harmful materials
Disease	Failure to maintain homeostasis

Organ system	Key function (per GR)
Skeletal	Support, protect soft tissues, mineral storage, blood cell formation, locomotion (with muscular)
Muscular	Locomotion, support, heat production
Nervous	Directs immediate responses to stimuli; coordinates other organ systems
Endocrine	Glandular system directing long-term changes in other organ systems' activity

Process	Definition
Cardiovascular	Internal transport of cells + dissolved materials (nutrients, wastes, gases)
Respiratory	Gas exchange between air and circulating blood
Integumentary, Lymphatic, Digestive, Urinary, Reproductive	(see Martini for full function lists — not directly quizzed beyond above in this GR set)

2. Cell Structure & Organelles

2.1 Membrane & surface structures

Structure	Description
Plasmalemma (cell membrane)	Phospholipid bilayer with embedded integral proteins + surface peripheral proteins
Integral proteins	Span the membrane; include gated channels, transporters, receptors
Peripheral proteins	Attached to membrane surface, not embedded through it
Glycocalyx	Carbohydrate coating on extracellular surface (glycoproteins/glycolipids) — cell identity/recognition
Sterols (cholesterol)	Stabilize membrane fluidity within the phospholipid bilayer
Microvilli	Apical surface projections that increase surface area for absorption
Cilia	Many short, hair-like motile structures per cell (vs. one flagellum); beat to move fluid/mucus across the surface
Flagellum	Whip-like motile structure; longer and typically one per cell (e.g., sperm tail) — distinct from cilia by length/number

2.2 Organelles

Organelle	Function
Mitochondria	ATP production (cellular respiration); have a DOUBLE membrane — smooth outer membrane + inner membrane folded into cristae — surrounding the matrix (interior fluid)
Golgi apparatus	Modifies/packages proteins; cis face receives from ER, trans face ships vesicles out; also renews/maintains the plasmalemma via vesicle trafficking, in addition to packaging/secretion
Rough ER	Ribosome-studded; synthesizes proteins for export/membranes
Smooth ER	Lipid synthesis, detoxification, calcium storage — no ribosomes
Ribosomes	Protein synthesis (free in cytosol or bound to rough ER)
Nucleolus	Site of ribosomal RNA synthesis/ribosome assembly within the nucleus
Lysosomes	Contain digestive enzymes; breakdown of debris, organelles, pathogens
Peroxisomes	Contain oxidases/catalase enzymes that break down fatty acids and toxins — e.g., neutralize hydrogen peroxide and alcohol
Centrioles	Organize microtubules; direct movement of chromosomes during mitosis
Centrosome	The region of cytoplasm containing the pair of centrioles; a centriole is one of the two structures within a centrosome

2.3 Cytoskeleton

Component	Role
Microtubules	Largest diameter; intracellular transport, form spindle apparatus & centrioles/cilia
Microfilaments (actin)	Smallest diameter; structural support, cell shape, contraction (thin filaments)
Intermediate filaments	Mid-sized; structural strength/stability (e.g., keratin)
Thick filaments	Myosin-based, specific to muscle cells (covered further in Section 16)

2.4 Nucleus

Structure	Description
Nuclear envelope	Double membrane surrounding the nucleus; contains nuclear pores
Nuclear pores	Allow exchange of materials between nucleus and cytosol
Nucleoplasm	Fluid contents of the nucleus
Chromatin	Loosely coiled DNA + histone proteins; condenses into chromosomes for division
Histones	Proteins around which DNA winds to form nucleosomes
Nucleosome	DNA wound around a histone core — basic packaging unit of chromatin

Cytosol vs. extracellular fluid: cytosol is high in K^+ and protein (low Na^+); extracellular fluid is high in Na^+ and low in protein (low K^+) — opposite gradients maintained by the Na^+/K^+ exchange pump.

3. Cell Junctions, Membrane Transport & Cell Cycle

3.1 Cell junctions

Junction type	Function
Gap junctions (communicating junctions)	Channel proteins connect adjacent cytoplasms — direct ion/small-molecule passage, electrical coupling; in cardiac muscle, gap junctions allow direct ion flow between adjacent cardiac muscle cells, enabling coordinated contraction
Zonula adherens (adhesion belt)	Encircles cell, anchored to microfilaments — mechanical stability
Hemidesmosomes	Anchor epithelial cells to underlying basal lamina
Desmosomes	Spot-welds linking adjacent cells via intermediate filaments
CAMs (cell adhesion molecules)	Transmembrane proteins that bind cells to each other and to extracellular matrix

3.2 Membrane transport mechanisms

Mechanism	Description
Diffusion	Passive movement of solute down its concentration gradient
Osmosis	Passive movement of water across a membrane toward higher solute concentration
Facilitated diffusion	Passive transport via carrier protein, down concentration gradient, no ATP
Active transport (exchange pumps)	ATP-powered movement against concentration gradient (e.g., Na^+/K^+ pump)
Endocytosis	Bringing extracellular material into the cell via vesicle formation
— Phagocytosis	"Cell eating" — engulfing large particles/pathogens
— Pinocytosis	"Cell drinking" — nonspecific uptake of extracellular fluid
— Receptor-mediated endocytosis/pinocytosis	Specific uptake via membrane receptor binding to target molecule — e.g., iron (transferrin-bound) and cholesterol (LDL-bound) are taken in this way
Exocytosis	Vesicle fuses with membrane to release contents outside the cell

3.3 Cell cycle & mitosis

Phase	Events
G0	Resting phase — cell performs normal functions, not preparing to divide
G1	Cell growth, organelle duplication, normal metabolic functions resume after division; G1 duration is variable — can be brief or very long depending on cell type
S	DNA replication — chromatin duplicated; DNA helicase unwinds the double helix, then DNA polymerase synthesizes the new complementary strand
G2	Final preparations for division (protein synthesis for mitotic apparatus)
Prophase	Chromatin condenses into visible chromosomes; spindle apparatus forms; nuclear envelope breaks down
Metaphase	Chromosomes align at the metaphase plate (cell's equator)
Anaphase	Sister chromatids separate, pulled to opposite poles by spindle fibers
Telophase	Nuclear envelopes re-form around each set of chromosomes; chromosomes decondense
Cytokinesis	Division of the cytoplasm, producing two daughter cells

Centrioles organize the mitotic spindle and direct chromosome movement during division. Interphase = G1 + S + G2 (the non-dividing portion of the cycle) — this is the phase in which a cell spends the MAJORITY of its life cycle.

4. Epithelial Tissue

4.1 General features

Feature	Description
Apical surface	Free/exposed surface of an epithelial cell (faces a lumen or exterior)
Basal surface	Surface attached to the basal lamina
Basolateral surface	Combined basal + lateral (side) surfaces
Basal lamina	Two-layered support structure under epithelium — dense layer (closer to epithelium) + clear layer; a layer of the basement membrane produced by the epithelium, composed of glycoproteins and proteoglycans, anchoring epithelium to underlying connective tissue

Feature	Description
Avascularity	Epithelium has no blood vessels — relies on diffusion from underlying connective tissue
Microvilli	Apical projections increasing absorptive surface area

4.2 Epithelial cell shapes & layering

Type	Description
Simple	Single layer of cells
Stratified	Multiple layers of cells
Squamous	Flat, scale-like cells
Cuboidal	Cube-shaped cells
Columnar	Tall, column-shaped cells
Pseudostratified columnar	Single layer, but cells of varying height give a layered appearance
Transitional epithelium	Stratified; cells can change shape (cuboidal ↔ squamous) to allow distention — found in urinary tract/bladder
Endothelium	Simple squamous epithelium lining blood vessels / heart chambers
Mesothelium	Simple squamous epithelium lining body cavities (serous membranes)

5. Glands

Term	Description
Exocrine gland	Secretes onto an epithelial surface, usually via a duct
Endocrine gland	Secretes hormones directly into the bloodstream (ductless); forms during development when the connecting duct/cells linking the gland to the epithelial surface disappear (losing their duct), unlike exocrine glands which retain one
Merocrine secretion	Vesicle contents released via exocytosis; cell remains intact (e.g., eccrine sweat glands)
Apocrine secretion	Loss of cytoplasm + secretory product pinched off at the apical surface
Holocrine secretion	Entire cell ruptures/disintegrates to release product (e.g., sebaceous glands)
Unicellular exocrine gland	Single secretory cell, e.g., goblet cell (mucus)
Multicellular exocrine gland — tubular	Secretory cells line a tube-shaped gland
Multicellular exocrine gland — acinar/alveolar	Secretory cells line a blind pocket (acinus)
Simple gland	Single duct connects to the surface
Compound gland	Duct branches repeatedly before reaching secretory cells
Simple coiled tubular gland	Merocrine secretion — example: eccrine sweat glands

6. Connective Tissue & Cartilage

6.1 Connective tissue cell types

Cell type	Function
Mesenchymal cells	Stem cells that can differentiate into other connective tissue cell types
Fibroblasts	Most abundant fixed cells; secrete collagen, elastic, and reticular fibers + ground substance
Fixed macrophages	Resident phagocytic cells, stay in a tissue
Free (wandering) macrophages	Mobile phagocytic cells that migrate through tissue
Mast cells	Release histamine/heparin during inflammation
Adipocytes	Fat storage cells
Eosinophils	Granular leukocytes that migrate into connective tissue during inflammation/parasitic response

6.2 Connective tissue fibers & ground substance

Component	Description
Collagen fibers	Triple-helix protein strands; high tensile strength, resist stretching
Elastic fibers	Contain elastin; stretch and recoil
Reticular fibers	Thin, branching collagen-based network providing structural support (e.g., in organs)
Ground substance	Gel-like medium surrounding cells/fibers, fills space between

6.3 Connective tissue proper — types

Type	Description
Loose (areolar) connective tissue	Open framework of fibers; fills space, surrounds vessels/nerves, cushions organs
Dense regular connective tissue	Parallel collagen fibers — tendons, aponeuroses, ligaments; resists pulling in one direction
Dense irregular connective tissue	Interwoven collagen fibers in multiple directions — resists stress from multiple directions (e.g., dermis, joint capsules)
Elastic tissue	Dominated by elastic fibers — allows stretch/recoil (e.g., some ligaments, large arteries)

6.4 Cartilage

Type / structure	Description
Hyaline cartilage	Resists compression; smooth, glassy matrix; found at joint surfaces, costal cartilage
Elastic cartilage	Contains elastic fibers — flexible, found in ear/epiglottis
Fibrocartilage	Dense collagen fiber matrix — found in intervertebral discs, menisci; resists compression + tension
Perichondrium	Fibrous covering around cartilage (except articular surfaces)
Chondrocytes	Mature cartilage cells, sit in lacunae
Chondroblasts	Immature cartilage-forming cells
Interstitial growth	Cartilage grows from within (chondrocytes dividing internally)
Appositional growth	Cartilage grows by adding new layers at the surface (from perichondrium) — the perichondrium contains stem cells (chondrogenic cells) that differentiate into chondroblasts, which add new cartilage matrix at the surface, as opposed to interstitial growth from within

Cartilage is avascular like epithelium; nutrients diffuse through the matrix, which is why it heals slowly.

7. Specialized Connective Tissue, Membranes, Fascia & Embryology

7.1 Fluid & supporting connective tissue

Type	Description
Blood	Fluid connective tissue — plasma + formed elements (cells)
Lymph	Fluid connective tissue circulating through the lymphatic system
Bone (osseous tissue)	Supporting connective tissue — calcified matrix (see Section 11)

7.2 Body membranes

Membrane	Description
Mucous membrane	Lines passages that communicate with the exterior (digestive, respiratory, urinary, reproductive tracts)
Serous membrane	Mesothelium + areolar tissue; has visceral layer (covers organ) + parietal layer (lines cavity wall); lines sealed internal cavities
Cutaneous membrane	The skin — thick, relatively waterproof, dry
Synovial membrane	Lines joint cavities; secretes lubricating synovial fluid; its lining cells are NOT true epithelium — they derive from fibroblasts and macrophage-like cells, not an epithelial lineage

7.3 Fascia layers

Layer	Description
Superficial fascia (hypodermis)	Loose connective tissue + adipose deep to dermis; connects skin to underlying structures
Deep fascia	Dense irregular connective tissue; divides muscles into functional compartments
Subserous fascia	Loose connective tissue between deep fascia and serous membranes lining body cavities

7.4 Muscle tissue types (overview — detail in Sections 16–18)

Type	Key features
Skeletal muscle	Multinucleated, striated, voluntary; myosatellite cells allow limited regeneration
Cardiac muscle	Striated, involuntary; intercalated discs join cells; pacesetter cells initiate contraction
Smooth muscle	Usually single nucleus, non-striated, involuntary; retains good regenerative capacity

7.5 Neural tissue

Component	Function
Soma (cell body)	Contains the nucleus and most organelles
Dendrites	Receive incoming signals
Axon (nerve fiber)	Conducts the outgoing signal away from the soma
Neuroglia	Support cells — provide structural framework + nutrients to neurons (do not conduct impulses)

7.6 Aging effects on tissue / embryology basics

Topic	Detail
Osteoporosis mechanism	Inactivity + low calcium intake + low estrogen contribute to excess bone loss vs. formation
Zygote	Fertilized egg — single cell formed at conception
Blastocyst	Early hollow ball-of-cells stage of the embryo
Trophoblast	Outer cell layer of blastocyst → becomes the placenta
Inner cell mass	Cluster within blastocyst → becomes the embryo itself

> *General aging principle (cross-system theme): aging tends to reduce tissue regenerative capacity, vascularity, and elasticity across multiple tissue/organ systems.*

8. Integumentary System — Epidermis

8.1 Epidermal layers (basal → surface)

Layer	Description
Stratum basale	Deepest layer, attached to basal lamina; site of continuous mitotic cell division; contains Merkel cells (touch) and melanocytes
Stratum spinosum	Several cell layers thick; keratinocytes connected by desmosomes; contains Langerhans (dendritic) cells
Stratum granulosum	Cells accumulate keratin precursors, begin to die
Stratum lucidum	Thin clear layer found ONLY in thick skin (palms/soles); poorly distinguishable in thin skin
Stratum corneum	Outermost; multiple layers of dead, keratinized cells; constantly shed and replaced

> *Cell turnover time from stratum basale to stratum corneum (shedding) is roughly 15–30 days. Among nucleated layers, the stratum spinosum is the most superficial layer that still has nucleated cells.*

8.2 Epidermal cell types

Cell type	Function
Keratinocytes	Most abundant epidermal cell — produce keratin, provide physical/water barrier
Tonofibrils	Bundles of intermediate filaments (keratin precursor protein bundles) within keratinocytes — provide structural support
Melanocytes	Produce melanin (pigment); located in stratum basale; melanin is stored in/transferred via membrane-bound vesicles called melanosomes
Langerhans (dendritic) cells	Phagocytic immune cells — found chiefly in stratum spinosum
Merkel cells	Sensory — stimulate associated sensory nerve endings for touch; found in stratum basale

8.3 Skin color & related signs

Factor	Description
Melanin	Pigment produced by melanocytes — main determinant of skin tone, protects against UV
Carotene	Orange-yellow pigment from diet, contributes to skin color (esp. with high intake)
Dermal blood supply	Oxygenated blood gives pink tone; contributes to overall skin color; DECREASES with age (T/F trick fact)
Cyanosis	Bluish skin discoloration from low oxygenation — caused by cold exposure or circulatory/respiratory disorders

9. Integumentary System — Dermis, Appendages & Repair

9.1 Dermis & hypodermis

Layer	Description
Papillary layer	Superficial dermis — loose (areolar) connective tissue, contains capillaries supplying epidermis

Layer	Description
Reticular layer	Deep dermis — dense irregular connective tissue, gives skin strength/elasticity
Cutaneous plexus	Network of blood vessels supplying the skin, located at dermis/hypodermis border
Hypodermis (superficial fascia)	Subcutaneous layer deep to dermis; stabilizes skin position relative to underlying tissue; stores fat (adipocytes)

9.2 Hair

Structure / term	Description
Hair follicle	Tube-shaped structure that produces a hair; includes the bulb at its base
Hair bulb	Base of the follicle; contains the papilla and matrix
Papilla	Dermal projection at the follicle base supplying blood to the matrix
Matrix	Region of dividing cells that produces the hair shaft
Hair shaft layers — cuticle / cortex / medulla	Cuticle = outer scaled layer; cortex = bulk of shaft; medulla = innermost core (may be absent in fine hair)
Vellus hair	Fine, pale "peach fuzz" hair
Terminal hair	Coarse, pigmented hair (scalp, eyebrows, after puberty: axilla/pubic)
Lanugo	Fine, soft hair covering the fetus
Arrector pili	Smooth muscle attached to follicle — contraction makes hair stand up ("goosebumps")
Folliculitis	Inflammation/infection of a hair follicle

9.3 Glands

Gland	Description
Merocrine (eccrine) sweat glands	Most widely distributed sweat gland; thermoregulation via sensible perspiration; myoepithelial cells contract to squeeze sweat into duct
Apocrine sweat glands	Found in axilla/groin; become active at puberty; secretion linked to pheromone production, can be odorous after bacterial action
Sebaceous glands	Holocrine secretion of sebum; usually associated with hair follicles
Ceruminous glands	Modified apocrine glands in ear canal — produce earwax (cerumen)
Mammary glands	Milk-producing glands, developmentally related to apocrine sweat glands

9.4 Nails & aging

Term	Description
Eponychium (cuticle)	Fold of skin overlapping the proximal nail body
Lunula	Pale crescent at the nail base (visible nail matrix)
Hyponychium	Area under the free edge of the nail body
Aging — Langerhans cell decline	Reduced immune sensitivity/surveillance in skin
Aging — decreased glandular activity	Reduced sweat/sebum output → overheating risk, drier skin
Aging — dermal thinning + elastic fiber loss	Wrinkling, reduced elasticity; wrinkle causes also include hormonal changes, UV radiation exposure, and pregnancy/weight fluctuation
Stem cells in skin (regeneration)	Stem/progenitor cells persist in BOTH the epidermis (epithelial, e.g. stratum basale) AND the dermis (connective tissue) throughout life, supporting regeneration of both layers

9.5 Wound healing & vitamin D

Step / topic	Description
1. Bleeding	Immediate response to injury, forms initial clot
2. Inflammatory response	Vasodilation, phagocyte recruitment to clean debris/pathogens
3. Granulation tissue	New capillary + fibroblast-rich tissue fills the wound
4. Collagen / scar formation	Fibroblasts deposit collagen, remodeling into mature scar tissue
Vitamin D (calcitriol)	Synthesized in skin with UV exposure; required for intestinal absorption of calcium and phosphorus

10. Bone Structure, Cells & Matrix

10.1 Gross bone regions

Region	Description
Diaphysis	Shaft of a long bone
Epiphysis	Expanded end(s) of a long bone
Metaphysis	Narrow zone connecting the epiphyseal cartilage (growth plate) to the diaphysis
Medullary cavity	Central marrow cavity within the diaphysis
Periosteum	Outer covering of bone — fibrous outer layer (protection/attachment) + osteogenic inner layer (cellular, bone growth/repair)
Endosteum	Incomplete cellular layer lining internal bone surfaces (medullary cavity, central canals); active in bone growth/repair/remodeling
Red marrow	Hematopoietic marrow — site of blood cell formation
Yellow marrow	Marrow dominated by adipocytes (fat cells) — contrast with red marrow (hematopoietic)

10.2 Compact vs. spongy bone microanatomy

Structure	Description
Osteon	Structural unit of compact bone — concentric lamellae around a central (Haversian) canal
Central (Haversian) canal	Contains blood vessels + nerves running through the long axis of the osteon
Perforating (Volkmann's) canals	Transverse canals carrying vessels/nerves from periosteum/endosteum to osteons deeper in the bone
Lamellae	Layers of bone matrix — concentric (around canal), circumferential (around whole bone), interstitial (remnants between osteons), radial
Canaliculi	Tiny channels connecting lacunae, allow nutrient/waste exchange between osteocytes
Lacunae	Small cavities housing osteocytes
Trabeculae	Interconnecting struts of spongy (cancellous) bone — no osteons, lighter, supports stress redistribution

10.3 Bone cells

Cell	Function
Osteoprogenitor cells	Stem cells; divide and differentiate into osteoblasts, e.g. after a fracture
Osteoblasts	Secrete the organic components of new bone matrix (osteoid)
Osteocytes	Mature bone cells; maintain the bone matrix; reside in lacunae
Osteoclasts	Multinucleated; dissolve bone matrix, release Ca^{2+} and phosphate (resorption)

10.4 Bone matrix composition

Component	Role
Organic matrix (collagen fibers)	Provides tensile strength / flexibility
Inorganic matrix (calcium phosphate / hydroxyapatite)	Provides hardness and resistance to compression

> *Periosteum, endosteum, the medullary cavity, and the epiphyses are all innervated by sensory nerves — bone is NOT insensate.*

11. Bone Surface Markings, Shapes & Vascular Supply

11.1 Surface marking terminology

Term	Meaning
Crest	Prominent ridge
Facet	Small, flat articular surface
Ramus	Angled extension/branch off the body of a bone
Trochlea	Smooth, pulley-shaped articular process
Sinus	Air-filled chamber within a bone
Head	Expanded articular end of a bone, often set off from the shaft by a neck
Fissure	Elongated cleft/slit
Condyle	Smooth, rounded articular process

Term	Meaning
Fossa	Shallow depression
Foramen	Opening for passage of nerve/vessel
Tuberosity / trochanter / tubercle	Prominent processes for tendon/ligament attachment (trochanter = largest, specific to femur)

11.2 Bone shape classification

Shape	Examples / description
Long bone	Shaft + 2 epiphyses (e.g., femur, humerus)
Short bone	Roughly cube-shaped (e.g., carpals, tarsals)
Flat bone	Thin, often curved plates; contain diploë (spongy bone) sandwiched between compact bone (e.g., skull, sternum, ribs)
Irregular bone	Complex shapes not fitting other categories (e.g., vertebrae)
Sesamoid bone	Small bone embedded within a tendon (e.g., patella)

11.3 Bone vascular supply

Vessel type	Supplies
Periosteal vessels	Compact bone of the shaft (via periosteum)
Nutrient vessels	Enter via nutrient foramen, supply diaphysis/medullary cavity
Metaphyseal vessels	Supply the metaphysis (near growth plate)
Epiphyseal vessels	Supply the epiphysis

12. Bone Growth, Fractures, Remodeling & Hormonal Regulation

12.1 Ossification

Type	Process
Intramembranous ossification	Bone forms directly from mesenchymal (or fibrous connective) tissue — e.g., flat skull bones, clavicle
Endochondral ossification	Bone forms by replacing a hyaline cartilage model; sequence: cartilage model enlarges → chondrocyte hypertrophy → cartilage calcifies → bone collar forms around diaphysis → osteoblasts invade and replace cartilage → primary ossification center (diaphysis) → secondary ossification centers (epiphyses)

12.2 Fracture types

Fracture type	Description
Transverse	Break straight across the shaft
Spiral	Caused by twisting stress
Comminuted	Bone shattered into 3+ pieces
Displaced	Bone fragments are misaligned
Compression	Bone crushed (common in vertebrae, porous bone)
Greenstick	Incomplete break — one side cracks, other bends (common in children)
Compound (open)	Bone breaks through the skin — infection risk

12.3 Fracture repair sequence

Step	Description
1. Hematoma formation	Blood clot forms at fracture site
2. Callus formation	External callus (fibrous + cartilaginous, from periosteum) + internal callus (from endosteum/marrow) bridge the fracture
3. Bony callus / remodeling	Callus replaced with bone, then remodeled to original shape over time

> *Bone remodeling is continuous throughout life — simultaneous deposition (osteoblasts) and resorption (osteoclasts); roughly a substantial fraction of the skeleton is recycled annually (estimates vary by source, ~1/5 to 1/3 per year).*

12.4 Hormonal regulation & disorders

Hormone / disorder	Effect
Growth hormone (GH)	Stimulates epiphyseal cartilage growth → long bone lengthening

Hormone / disorder	Effect
Sex hormones (estrogen/testosterone)	Accelerate epiphyseal closure; contribute to male/female size + shape differences
Calcitonin	Lowers blood calcium — stimulates osteoblast activity / inhibits osteoclasts
Parathyroid hormone (PTH)	Raises blood calcium — stimulates osteoclast activity, increases renal calcium retention
Calcitriol (active vitamin D)	Increases intestinal absorption of calcium and phosphate
Achondroplasia	Disproportionate short stature — slowed epiphyseal cartilage growth/replacement
Osteoporosis	Excess bone resorption vs. formation; most common in older women (low estrogen, inactivity, low calcium intake)
Gigantism	Excess growth hormone before epiphyseal closure → excessive long bone growth
Marfan's syndrome	Connective tissue disorder; abnormally long limbs (common distractor vs. gigantism)
Osteomalacia	Adult bone softening from vitamin D deficiency / poor mineralization
Rickets	Childhood version of osteomalacia; impairs bone growth

12.5 Skeletal system functions (summary)

Support; protection of soft tissues/organs; mineral storage (calcium, phosphate) and homeostasis; blood cell formation (hematopoiesis) in red marrow; leverage for movement (with muscles); shock absorption.

Age-related decline in bone mass typically begins around age 30-40.

12.6 Bone & joint diagnostic procedures

Procedure	Use
Arthrocentesis	Joint fluid aspiration for diagnosis
DEXA scan	Bone density measurement
Plain X-ray	Standard imaging for bone/joint structure and fractures

13. Skeletal Muscle — Organization, Connective Tissue Layers & Naming

The human body has approximately 700 skeletal muscles.

13.1 Connective tissue layers (outside → inside)

Layer	Description
Epimysium	Dense irregular connective tissue surrounding the ENTIRE muscle; continuous with deep fascia
Perimysium	Surrounds each FASCICLE (bundle of muscle fibers); carries vessels/nerves into the muscle
Endomysium	Surrounds each individual MUSCLE FIBER; contains myosatellite cells, capillaries
Fascicle	A bundle of muscle fibers within the muscle, wrapped by perimysium
Myosatellite cells	Stem cells between endomysium and muscle fibers; aid in repair of damaged muscle tissue

Order from inside to outside: endomysium → perimysium → epimysium. Epimysium connects to deep fascia (not superficial fascia).

13.2 Muscle fiber arrangements

Arrangement	Description
Parallel	Fibers run parallel to long axis; most common pattern
Convergent	Fibers spread over a broad area but converge onto a single common attachment/tendon
Pennate (uni-/bi-/multi-)	Fibers at an angle to the tendon, like a feather — generates the MOST tension/power of any arrangement
Circular (sphincter)	Fibers concentrically arranged around an opening — surrounds entrances to digestive/urinary tracts

13.3 Muscle attachments, levers & naming

Term	Description
Origin	Proximal/fixed attachment point of a muscle (typically moves less)
Insertion	Distal/mobile attachment point (typically moves more); insertion is typically distal to origin
Tendon	Cord-like attachment of muscle to bone
Aponeurosis	Broad, flat sheet-like tendon attachment
First-class lever	Fulcrum lies BETWEEN the applied force and the resistance (e.g., head/neck)
Second-class lever	Load lies BETWEEN the fulcrum and the applied force
Third-class lever	Applied force lies between the fulcrum and the load (most common in the body)

Term	Description
Lateralis (naming)	Refers to side of the body/structure
Latissimus (naming)	Refers to width — "widest"
Longissimus (naming)	Refers to fiber orientation along the long axis of the body

13.4 Muscle action roles & motor units

Term	Definition
Agonist (prime mover)	Muscle mainly responsible for producing a given movement
Synergist	Muscle that assists the agonist
Antagonist	Muscle that opposes the action of the agonist
Fixator	Stabilizes a joint/bone so the agonist can act effectively
Motor unit	One motor neuron + all the muscle fibers it controls; one action potential affects one motor unit
Motor unit size	Varies widely — large leg muscles can have many thousands of fibers per motor unit; smaller/precise muscles (eyes, fingers) have far fewer
Recruitment	Increasing the NUMBER of motor units activated — how the nervous system controls force precisely
Muscle tone	Continuous, low-level tension in a resting muscle without producing visible movement
Hypertrophy	Increase in muscle fiber size from repeated near-maximal stimulation
Atrophy	Decrease in muscle size/fiber diameter from disuse (e.g., a limb in a cast)

14. Sarcomere Structure & Contraction Physiology

14.1 Sarcomere bands & zones

Structure	Description
Z line (Z disc)	Marks the END of each sarcomere; composed of interconnecting proteins (actinin) anchoring thin filaments — NOT thick filaments
M line	Midline of the sarcomere; anchors thick filaments at the center
A band	Dark band — contains the FULL LENGTH of thick filaments, including the zone of overlap
I band	Light band — region with thin filaments only, no thick filaments
H band	Region within the A band with thick filaments only, no overlap with thin filaments
Zone of overlap	Region where thick and thin filaments interdigitate; each thin filament surrounded by 3 thick filaments, each thick filament surrounded by 6 thin filaments
Actinin	Forms the open meshwork of the Z line, anchoring thin filaments

Polarized light: A bands = dark; I bands = light. The A band stays constant length during contraction; the H band and I band shrink/disappear as the Z lines are pulled closer together.

14.2 Filament composition

Filament / molecule	Description
Thick filament (myosin)	Double-stranded myosin molecules with elongated tails + free globular heads (cross-bridges) that bind actin
Thin filament (actin)	F-actin strands; troponin-tropomyosin complex regulates access to myosin-binding active sites
Nebulin	Accessory protein running along thin filaments; helps organize/anchor actin
Titin	Elastic accessory protein connecting Z line to the thick filament/M line; provides elastic recoil
Troponin	Binds calcium ions; binding causes a shift in the troponin-tropomyosin complex, exposing actin active sites
Tropomyosin	Covers myosin-binding active sites on actin at rest; pivots aside when troponin binds Ca^{2+}
Myofibril	Bundle of thick + thin filaments organized into repeating sarcomeres — the contractile element of a muscle fiber
Sarcoplasmic reticulum (SR)	Smooth-ER-like membrane network storing and releasing calcium ions
Triad	One T tubule + two terminal cisternae of the SR — couples electrical signal to calcium release
T tubule (transverse tubule)	Extension of the sarcolemma into the cell — distributes the action potential, triggers Ca^{2+} release from SR

14.3 Neuromuscular junction & excitation–contraction coupling

Step / structure	Description
Synaptic terminal	Expanded axon tip facing the motor end plate; releases acetylcholine (ACh)

Step / structure	Description
Synaptic vesicle	Small membrane-bound sac within the synaptic terminal that stores/releases neurotransmitter — NOT the same as the synaptic cleft (commonly confused on T/F questions)
Synaptic cleft	The GAP between the neuron and muscle fiber, separating the synaptic terminal from the motor end plate — distinct from the synaptic vesicle
Motor end plate	Specialized region of sarcolemma with ACh receptors, opposite the synaptic terminal
Acetylcholine (ACh)	Neurotransmitter released by the motor neuron; binds receptors on motor end plate
1. ACh binds receptors	Triggers an action potential in the sarcolemma immediately
2. Action potential spreads via T tubules	Triggers Ca^{2+} release from the sarcoplasmic reticulum (triad)
3. Ca^{2+} binds troponin	Troponin-tropomyosin complex shifts, exposing actin active sites
4. Cross-bridge formation	Myosin heads bind exposed actin active sites
5. Power stroke	Myosin heads pivot toward the M line, pulling thin filaments inward (sliding filament mechanism) — sarcomere shortens
6. Relaxation	Acetylcholinesterase (AChE) breaks down ACh in the synaptic cleft, ending stimulation; Ca^{2+} is pumped back into SR

Calcium is the direct trigger for contraction (binds troponin). The sliding filament mechanism = myosin heads bind actin active sites and pivot toward the M line. ATP hydrolysis powers cross-bridge cycling and is also required for relaxation (detachment).

14.4 Sarcomere length & tension

Sarcomere state	% maximum tension produced
Optimal resting length (~2.0–2.25 µm)	100% (optimal)
Highly stretched (well above ~3.1 µm)	Fairly weak (~20–40%) — too little filament overlap

15. Muscle Fiber Types, Metabolism & Motor Control

Fiber type	Key characteristics
Fast (white) fibers	Largest diameter; densely packed myofibrils; large glycogen reserves; few mitochondria; most numerous fiber type overall; favor anaerobic/glycolytic metabolism; good for sprinters
Slow (red) fibers	Smaller diameter (~2/3 of fast fiber); high myoglobin content (red color); more mitochondria, better blood supply; use aerobic metabolism (carbohydrates, lipids, amino acids); fatigue-resistant; take ~2x longer to contract after stimulation than fast fibers
Intermediate fibers	Properties between fast and slow; fast contraction speed but more myoglobin/mitochondria + fatigue resistance than typical fast fibers

Myoglobin is the oxygen-binding pigment responsible for the red color of slow fibers. A very good sprinter likely has more fast-twitch fibers than average; aging muscle develops increasing fibrous connective tissue.

15.2 Tension summation & control

Term	Description
Twitch	Single contraction-relaxation cycle from one stimulus
Summation	Repeated stimulation before full relaxation → increasing tension
Peak tension (tetanus)	Maximal sustained tension when all motor units fire at maximal rate
Recruitment	Increasing the number of motor units activated to increase force
Precision of control	A motor unit with FEWER fibers per neuron allows MORE precise control (e.g., eye muscles); more fibers per unit = less precise but more powerful

16. Skull Facts, Skeletal Aging & Joints

16.1 Skull & related structures

Structure	Description
Paranasal sinuses	Air-filled chambers connected to nasal cavity; humidify/warm air, resonate during sound production, lighten skull bones
Auditory ossicles	Tiny bones within the temporal bone (malleus, incus, stapes) that transmit sound for hearing
Styloid process	Temporal bone projection — attachment point for muscles that extend/rotate the head
Facial bones	Frontal, sphenoid, ethmoid, nasal, lacrimal, zygomatic, maxillae, mandible, etc. — protect/support entrances to digestive + respiratory tracts

16.2 Skeletal aging & forensic features

Feature	Detail
Epiphyseal cartilage fusion	Begins around age 18 (varies by bone)
Fontanel closure	Typically complete by about age two
Bone mineral content with age	Mineral content diminishes as a normal part of aging (matrix composition is NOT constant through life)
Age estimation from skeleton	Degree of ossification of cranial sutures is a key indicator
Skeleton can reveal	Forensic anthropologists CAN estimate age, sex, body size, AND race/ancestry from skeletal features
Female pelvic/skeletal traits	Generally lighter bone, smoother/smaller prominences, heart-shaped pelvic inlet, wider pelvic outlet
Child vs. adult skull	Frontal suture present until ~age 2–8; full dentition absent; smaller mastoid process; proportionally larger cranium

16.3 Joint classification

Class	Mobility / example
Synarthrosis	Immovable joint (e.g., suture)
Amphiarthrosis	Slightly movable joint (e.g., symphysis, syndesmosis)
Diarthrosis (synovial joint)	Freely movable joint with a joint cavity
Synchondrosis	Cartilaginous joint, generally immovable (e.g., epiphyseal plate)
Symphysis	Cartilaginous joint permitting slight movement (fibrocartilage pad, e.g., pubic symphysis)
Synostosis	Joint completely fused into a single bone
Gomphosis	Fibrous joint anchoring a tooth in its socket
Pivot joint	Allows rotational movement only (e.g., atlantoaxial joint)

16.4 Joint accessory structures

Structure	Function
Joint (articular) capsule	Fibrous structure enclosing a synovial (diarthrotic) joint
Synovial membrane	Lines the joint capsule (except articular cartilage), secretes synovial fluid
Articular cartilage	Hyaline cartilage covering bone ends at the joint surface
Meniscus	Fibrocartilage pad that subdivides a synovial cavity / channels synovial fluid flow / allows variation in articulating surface shape
Bursa	Fluid-filled sac reducing friction, fills space from changing joint shape, packing material at joint periphery
Fat pad	Packing material around joint periphery
Ligament	Connects bone to bone, stabilizes joint

16.5 Joint movements

Movement	Description
Flexion	Decreases the angle between articulating elements (anterior-posterior plane)
Extension	Increases the angle between articulating elements
Abduction	Movement AWAY from the longitudinal axis/midline, in the frontal plane
Adduction	Movement TOWARD the longitudinal axis/midline
Rotation	Turning movement around an axis
Circumduction	Circular movement combining flexion/extension/abduction/adduction
Protraction / retraction	Forward / backward movement in a horizontal plane
Depression / elevation	Downward / upward movement
Eversion / inversion	Turning the sole of the foot outward / inward
Opposition	Thumb movement to touch the fingertips

16.6 Joint disorders

Term	Description
Arthritis	Always involves damage to the articular cartilage (joint inflammation/degeneration)
Luxation (dislocation)	Articulating surfaces forced entirely out of position
Subluxation	Partial dislocation